

Helix Thready — MVP Documentation Integration Report

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Rev	Date	Author	Change
1	2026-07-21	orchestrator (integration)	Initial integration report — tree, per-area file counts, cross-reference verification, open-item & gap register roll-up

This report records the **integration pass** over the eight MVP documentation areas under `docs/public/research/mvp/`. It verifies that every cross-area reference resolves, that each area index carries an Upstream/Downstream note consistent with the **master dependency map**, and it rolls up the remaining `[OPEN: ...]` items and gap-register coverage across all areas. All content follows **CONVENTIONS.md**.

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1. Summary

- **8 areas, 74 Markdown docs, 84 Mermaid `.mmd` diagram sources, and 4 machine artifacts** (`api/openapi.yaml` , `database/schema-relational.sql` , `database/schema-vector.sql` , `database/migrations/0001_init.sql`) — **162 files** total (incl. the root `index.md` , `CONVENTIONS.md` , and this report is the 163rd once written).
- **Cross-references: PASS.** 736 file-targeted Markdown links checked; **0 broken**. All 110 cross-parent links resolve to real files (the lone regex hit was an inline-code *example* of the link syntax in `CONVENTIONS.md §5` , not a rendered link).
- **Status tracker updated** in `index.md`: Pass 1 (draft) + Pass 2 (review) + Integrated marked done for all eight areas; Pass 3 (optional polish) left open.
- **Every area index carries an Upstream/Downstream note** and is directionally consistent with the master dependency map (see §5).
- **132 unique `[OPEN: ...]` markers** and **82 `ATM-*` workable-item ids** are tracked; **178 `[GAP: ...]` tags** address the P0/P1/P2 register — none papered over.

2. Documentation tree

```

docs/public/research/mvp/
├─ index.md                      (master index & roadmap – status tracker)
├─ CONVENTIONS.md                (documentation conventions)
├─ INTEGRATION_REPORT.md         (this file)
├─ architecture/
│   └─ index.md
│   └─ system-overview.md · component-catalog.md · data-flow.md · event-model.md
│   └─ concurrency-and-idempotency.md · service-discovery.md · security-model.md
│   └─ messenger-ingestion.md · processing-pipeline.md · semantic-search.md · asset-and-
download.md
├─ diagrams/                      (15 .mmd)
├─ api/
│   └─ index.md
│   └─ openapi.yaml              (OpenAPI 3.1 – the REST /v1 contract)
│   └─ rest-endpoints.md · event-bus-contract.md · authn-authz.md · error-model.md
│   └─ versioning.md · sdk-strategy.md · contract-tests.md
│   └─ diagrams/                  (7 .mmd)
├─ database/
│   └─ index.md · erd.md · indexing.md · partitioning.md · retention-archive.md · migration-
strategy.md
│   └─ schema-relational.sql · schema-vector.sql
│   └─ migrations/0001_init.sql
│   └─ diagrams/                  (10 .mmd)
├─ deployment/
│   └─ index.md
│   └─ container-topology.md · podman-compose.md · environments.md · tls-lets-encrypt.md
│   └─ deploy-and-rollback.md · backup-dr.md · service-discovery-ports.md
│   └─ hetzner-provisioning.md · secrets-and-config.md
│   └─ diagrams/                  (10 .mmd)
├─ development/
│   └─ index.md · workable-items.md · agent-orchestration.md · submodule-map.md
│   └─ coding-standards.md · contribution-guidelines.md · build-new-subsystems.md
│   └─ diagrams/                  (9 .mmd)
├─ testing/
│   └─ index.md · test-strategy.md · test-types.md · tdd-skeletons.md · helixqa-banks.md
│   └─ challenges-scenarios.md · static-analysis.md · performance-and-chaos.md
│   └─ diagrams/                  (8 .mmd)
├─ design/
│   └─ index.md · design-system.md · brand-assets.md · theming.md · wireframes.md
│   └─ ux-flows.md · component-library.md · prototypes.md
│   └─ diagrams/                  (13 .mmd)
├─ user-guides/
│   └─ index.md · installation.md · configuration.md · root-admin-guide.md · account-admin-
guide.md
│   └─ end-user-manual.md · cli-reference.md · tui-usage.md · web-portal-guide.md · mobile-
guide.md
├─ sdk-quickstart.md · faq.md · troubleshooting.md
└─ diagrams/                      (12 .mmd)

```

3. Per-area file counts

Area	Markdown	<code>.mmd</code> diagrams	Other	Area total
architecture	12	15	0	27
api	8	7	1 (<code>openapi.yaml</code>)	16
database	6	10	3 (2× <code>.sql</code> + 1 migration)	19
deployment	10	10	0	20
development	7	9	0	16
testing	8	8	0	16
design	8	13	0	21
user-guides	13	12	0	25
root (<code>index.md</code> , <code>CONVENTIONS.md</code>)	2	0	0	2
Total	74	84	4	162

Every embedded Mermaid diagram has a sibling `.mmd` source under the area's `diagrams/` folder per `CONVENTIONS.md` §4; the `.mmd` count (84) exceeds the doc count because several docs embed multiple diagrams.

4. Cross-references resolved

A full link sweep parsed every Markdown link in all 74 docs and resolved the file portion of each target relative to its source file.

Metric	Result
File-targeted Markdown links checked	736
Cross-parent (<code>../...</code>) links	110
Broken links	0
Cross-area file-specific links (non- <code>index.md</code>)	<code>design/theming.md</code> → <code>../database/erd.md</code> (×2) — resolves

- Every area `index.md` links to its own files **and** to the sibling area indexes it depends on (`CONVENTIONS.md` §5); all eight sibling `index.md` targets exist and resolve.
- The private authoritative sources are referenced with correct relative depth: `../../../../private/...` from the root `index.md` / `CONVENTIONS.md` , and `../../../../private/...` from the area subdirectories (both verified against the `docs/private/research/mvp/` location).

- The single link-checker hit (`CONVENTIONS.md → ../architecture/event-model.md`) is an inline-code **example** of the relative-link convention (wrapped in backticks, never rendered), not an actual reference. No fix required.

5. Upstream/Downstream reconciliation with the master map

The master dependency map draws these contract edges ($A \rightarrow B \equiv$ “A feeds B”, A upstream):

`architecture → {api, database, deployment, testing}`, `database → {api, testing}`, `api → {user-guides, design, testing}`, `design → user-guides`, and `development → (underpins every area)`.

All eight area indexes carry an explicit **Upstream/Downstream** note, and every one is directionally consistent with the map:

Area	Upstream note lists	Downstream note lists	Consistent with master map
architecture	authoritative sources, in-house engines (root of the DAG)	api, database, deployment, testing	✓
api	architecture, database, in-house modules	design, user-guides, testing	✓
database	architecture, research, in-house modules	api, deployment, testing, development	✓
deployment	architecture, database, api	development, testing	✓
development	architecture, api, database, decision-matrix + gap register	testing, deployment, design, user-guides	✓ (process/ underpinning axis)
testing	architecture, api, database	deployment, development	✓
design	architecture, api, <code>design_system</code> / OpenDesign, sources	user-guides, testing, impl repos	✓
user-guides	architecture, api, database, deployment, design	testing, docs-chain, operators	✓

Note on `development`. The master map renders `development → (all)` because development is the cross-cutting **process/underpinning** area — it defines the workable items and standards that *produce* each section. The development index additionally lists architecture/api/database as *upstream* because it **consumes their contracts** to author those items. These are two axes (process vs. contract) of the same

relationship, not a contradiction; the map’s own prose calls development the layer that “underpins everything.” Its bidirectional edges with testing and deployment (dev feeds the build backlog; those areas feed requirements back) are intentional and documented in each index.

6. Remaining open items across all areas

132 unique `[OPEN: ...]` **markers** remain tracked (each tied to an `ATM-*` /plan, none papered over). The headline register per area index:

Area	Index-level open items (headline)
architecture	<code>OVERVIEW-1/2</code> , <code>CAT-1/2</code> , <code>EVT-1/2</code> , <code>CONC-1/2</code> , <code>DISC-1/2</code> , <code>SEC-1/2/3</code> , <code>ING-1/2/3</code> , <code>PROC-1/2/3/4</code> , <code>SEM-1/2/3</code> , <code>ASSET-1/2/3</code> , <code>DF-1/2/3</code> (source-verification of FLAGGED interfaces)
api	<code>api-1</code> (proto files produced with code), <code>api-2</code> (rate-limit tiers pending billing), <code>api-3</code> (Zig hand-written SDK)
database	<code>ATM-DB-001/002/004/011/012/013/021/031/032/033</code> (pg placeholders, CONCURRENTLY, partition FK, multi-lang FTS, pgvector-vs-Qdrant benchmark, vector tenant isolation, GDPR cold erasure, MinIO signed-URL parity, sub-partitioning)
deployment	<code>dns-provider</code> , <code>host-sizing</code> , <code>secondary-store</code> , <code>buildnew-images</code>
development	<code>constitution-anchor-verify</code> (<code>ATM-066</code>), <code>max-oneme-go-port</code> (<code>ATM-018</code>), <code>agentpool-contract</code> (<code>ATM-067</code>); <code>sibling-area-index-missing</code> (<code>ATM-072</code>) — CLOSED this pass
testing	<code>canonical-helixqa-repo</code> , <code>helixstream-scope</code> , <code>docs-chain-tooling</code> , <code>mobile-device-farm</code>
design	<code>THREADY-DES-01</code> (Logo eyedrop), <code>THREADY-DES-02</code> (PenPot/Lottie bridge), <code>THREADY-DES-03</code> (heart glyph color), <code>THREADY-DES-14</code> (branding storage+contract reconciliation with database/api)
user-guides	per-guide open items + the <code>docs_chain</code> honest-SKIP note (<code>[GAP: 19]</code>); endpoint/entity names track the api/database areas

Cross-area open items worth tracking to closure at the program level:

- **`THREADY-DES-14`** (design ↔ database ↔ api): the `account_branding` normalized table vs the canonical `accounts.branding` JSONB, and the expanded `setBranding` contract vs `api/openapi.yaml` — must converge on the canonical schema/contract; design MUST NOT diverge.
- **`ATM-DB-033`** (database ↔ deployment): MinIO signed-URL parity — a storage/deployment concern the schema defers to via an opaque `storage_key`.
- **`docs-chain-tooling`** (testing ↔ deployment ↔ user-guides): `pandoc` / `weasyprint` provisioning gates md→HTML/PDF/PNG sibling generation; the Markdown remains the source of truth meanwhile.

7. Gap-register coverage

178 `[GAP: ...]` tags across the tree map the P0/P1/P2 items from

`helix_thready_subsystem_gaps_and_improvements.md` onto owning documents. The P0 traps are each owned by a design plan or `[BUILD-NEW]` item — never claimed to “work”:

P0 gap	Owning area(s)
<code>#1 / 2.1</code> HelixLLM <code>HashEmbedder</code> non-semantic default	architecture (semantic-search), api, database, deployment, user-guides
<code>#2 / 2.6</code> VisionEngine has no OCR engine	architecture (processing-pipeline), user-guides
<code>#6 / 4.1</code> helix_skills has no execution engine	architecture (processing-pipeline), api, database, user-guides
<code>#3 / 5.1</code> Herald Telegram in QA harness; Max empty stub	architecture (messenger-ingestion), api, user-guides
<code>6.2 / #4</code> filesystem: no HTTP source / download semantics	architecture (asset-and-download), user-guides
<code>6.3</code> Download Manager does not exist	architecture (asset-and-download), development (build-new), user-guides
<code>6.5 / #5</code> MeTube poll-only, no outbound webhook	architecture (asset-and-download), api, user-guides
<code>7.3</code> Security-KMP mobile secure storage is a stub	architecture (security-model), design, user-guides

P1/P2 items (JWT asymmetric keys, partitioning helpers, LLMsVerifier port `:7061` vs `:8080`, Boba callback contract, standardized callback/task module, searchable-sealed credentials, VisualRegression CI, docs_chain pandoc/weasyprint) are each addressed in the owning area’s gap table with a `[GAP: ...]` tag and a tracked plan. The BUILD-NEW subsystem set (Asset/Download/User/Max/OCR/webhook/ callback/ EventBus service/ThreadReader/Semantic-search) is declared as placeholders in `deployment/container-topology.md` and designed in `development/build-new-subsystems.md`.

8. Changes applied in this pass

- index.md** — status tracker: Pass 1, Pass 2 and Integrated marked done (☑) for all eight areas; added a rev-2 row and a link to this report.
- development/index.md** — closed `[OPEN: sibling-area-index-missing]` / `ATM-072`: `architecture/index.md` and `database/index.md` are present and the §3 cross-links resolve; added a rev-3 row.
- user-guides/index.md** — updated the §4 cross-area caveat: all upstream sibling indexes are committed and every link resolves; added a rev-2 row.

4. **INTEGRATION_REPORT.md** — this file (new).

No broken links required fixing; no Upstream/Downstream note required a directional correction. The edits above reconcile the two now-stale “sibling index not yet committed / link may 404” notices that predated area integration.

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